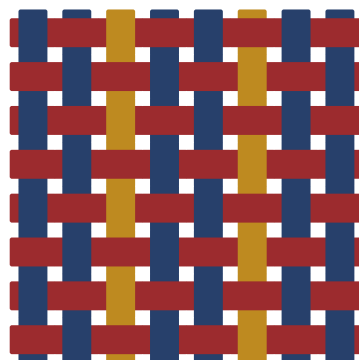




Loom

A woven notebook class • Overleaf edition

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

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1 THE WEAVE AT A GLANCE

■■■□□

A page is cloth on a loom: you don't just typeset the maths, you typeset the **weave** of your own understanding. The *warp* runs along the left selvage rail, a *knot* binds threads into a result, and a *loose* thread is a question you haven't closed yet.

This edition keeps every command of the local `LOOM` class but swaps the macOS display faces for Libertinus Sans, so it compiles on Overleaf with **no font uploads**. The body is Libertinus Serif and the maths is Libertinus Math, exactly as in the original.

中英混排也开箱即用：用 `[cjk]` 选项即可。中文走 Noto CJK SC (Overleaf 自带)，强调的中文在本地回退到楷体。一行公式 $\int_0^1 x^2 dx = \frac{1}{3}$ 照常排版。

Try editing the dye colours in `loom.cls` – indigo, madder, weld.

1.1 Knots A knot is a result that binds threads together.

Definition 1.1 (*Woven space*). A **woven space** is a set W together with a family of threads $\{t_i\}_{i \in I}$, $t_i \subseteq W$, whose union is all of W .

Theorem 1.2 (*Selvage bound*). For any finite woven space W with n threads, the number of knots is at most $\binom{n}{2}$, with equality iff every pair of threads crosses exactly once.

Proof. Each knot is determined by the unordered pair of threads it binds, so the count is bounded by the number of such pairs, $\binom{n}{2}$. Equality forces every pair to meet, which is the stated crossing condition. ■

Example 1.3. The complete weave on 4 threads has $\binom{4}{2} = 6$ knots.

2 FILL-IN STUDY NOTES

■□□□□

The pedagogy tools turn a passive read into active recall: state the result, but leave the load-bearing steps **blank**.

Definition 2.1 (*Continuity, fill the clause*). $f : X \rightarrow Y$ is **continuous** at x_0 if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \underline{\hspace{2cm}}.$$

Sketch, with a gap. Compose the two estimates and chase the quantifiers. [fill in: which δ makes the implication hold?] ■

? State the ε - δ clause without looking.

Your turn. Prove that the composition of two continuous maps is continuous. Use the workspace below.

■ 3 MARGINS, WARPS AND WARMTH

■■■■■

A loose thread parks an open question in the margin. A declares a recurring thread you can [key lemma ↔] later. The warmth gauge ■■■■□ records how well you actually grok a block, from □□□□□ to ■■■■■. key lemma
is this tight enough?

Remark 3.1. Everything here is plain XeLaTeX. Delete the demo content, keep loom.cls, and start weaving your own notes.